System Architecture & Agent Flowchart

NetElixir Revenue Forecaster — AIgnition 3.0

Track: Track B (Multi-Agent AI & Interactive Dashboard) **Status:** Production Ready

1. Architectural Overview

The NetElixir Revenue Forecaster is engineered as a highly modular, production-grade predictive pipeline combined seamlessly with an interactive decision-support layer. The system is intentionally decoupled into discrete functional components. This structural separation guarantees exceptional scale properties, bulletproof fault tolerance, and absolute reliability during automated grading executions within an inference-only sandbox environment.

2. Core Functional Components

- **Data Ingestion & Alignment Layer:** Programmatically scans, validates, and parses cross-channel historical advertising metrics extracted from three independent data environments: Google Ads, Meta Ads, and Bing Ads.
- **Feature Engineering Engine (src/generate_features.py):** Ingests raw multi-platform structures, standardizes variable naming schemas, handles structural anomalies or missing flags, performs normalization scaling, and constructs cross-channel performance interactions.
- **High-Speed Serialization Layer (src/processed_features.parquet):** Rather than exporting intermediate structures into costly uncompressed text files, features are committed directly via Apache Parquet binary compression. This slashes pipeline I/O delays, strictly locks system datatypes, and protects memory layout constraints.
- **Inference Engine (src/predict.py):** Automatically triggers immediately upon feature availability. It maps live vectors directly against the serialized, pre-trained model binary (pickle/model.pkl) to calculate high-precision revenue matrix forecasts, committing the target directly into output/predictions.csv.
- **Interactive Simulation UI Layer (app.py):** The client face built on a customized Streamlit framework. It enables end-users to dynamically simulate marketing budget reallocations across platforms using responsive graphical panels and real-time inference updating.
- **AI Synthesis Insights Agent:** Operating entirely via environment-secured operational variables (.env), this component feeds live forecasting outputs directly into an LLM synthesis API engine to automatically write rich executive analytical briefs based on simulation movements.

> **Design Paradigm Note:** The entire pipeline architecture operates on a deterministic execution flow for data preprocessing and inference steps, utilizing generative models exclusively for summary visualization and context building, eliminating hallucination risks in the math core.

3. Execution Data Flow Diagram

```
[Raw CSV Data (Google / Meta / Bing)]
               │
               ▼
[Feature Engineering Engine (src/generate_features.py)]
               │
               ▼
[High-Speed Parquet Storage Layer (src/processed_features.parquet)]
               │
               ▼
[Inference Engine (src/predict.py)] <─── [Pre-trained Model (pickle/model.pkl)]
               │
               ├───> Outputs directly to Grader Sandbox ───> [output/predictions.csv]
               │
               ▼
[Interactive Streamlit Dashboard UI (app.py)]
               │
               ▼
[AI Synthesis Insights Agent (.env Key Access)] ───> [Automated Executive Summary]
```